

FRONTEND

Frontend Framework: React (with functional components and hooks)

Styling: Tailwind CSS or plain CSS

- Project Structure (Simplified)

```
quiz-app/  
├── src/  
│   ├── components/  
│   │   ├── Quiz.js  
│   │   ├── Question.js  
│   │   └── Result.js  
│   ├── data/  
│   │   └── questions.js  
│   ├── App.js  
│   └── index.js  
└── package.json
```

- DEFINE QUIZ DATA (questions.js)

```
const questions = [  
  {  
    question: "What is the capital of France?",  
    options: ["Berlin", "Madrid", "Paris", "Rome"],  
    answer: "Paris"  
  },  
  {  
    question: "Who developed React?",  
    options: ["Facebook", "Google", "Twitter", "Microsoft"],  
    answer: "Facebook"  
  },  
  {  
    question: "What does HTML stand for?",  
    options: [  
      "Hyper Trainer Marking Language",  
      "Hyper Text Markup Language",  
      "Hyper Text Marketing Language",  
      "Hyper Text Markup Leveler"  
    ],  
    answer: "Hyper Text Markup Language"  
  }  
];
```

```
export default questions;
```

- APP LOGIC(App.js)

```
const questions = [  
  {  
    question: "What is the capital of France?",  
    options: ["Berlin", "Madrid", "Paris", "Rome"],  
    answer: "Paris"  
  },  
  {  
    question: "Who developed React?",  
    options: ["Facebook", "Google", "Twitter", "Microsoft"],  
    answer: "Facebook"  
  },  
  {  
    question: "What does HTML stand for?",  
    options: [  
      "Hyper Trainer Marking Language",  
      "Hyper Text Markup Language",  
      "Hyper Text Marketing Language",  
      "Hyper Text Markup Leveler"  
    ],  
    answer: "Hyper Text Markup Language"  
  }  
];
```

```
export default questions;
```

- DISPLAY QUESTIONS AND OPTIONS(Quiz.js)

```
import React, { useState } from "react";
```

```
const Quiz = ({ questionData, questionNumber, totalQuestions, onAnswer }) => {  
  const [selected, setSelected] = useState("");
```

```
  const handleSubmit = () => {  
    if (selected) {  
      onAnswer(selected);  
      setSelected("");  
    }  
  };  
};
```

```
  return (
```

```

<div className="quiz">
  <p>
    Question {questionNumber}/{totalQuestions}
  </p>
  <h3>{questionData.question}</h3>
  <ul>
    {questionData.options.map((opt, index) => (
      <li key={index}>
        <label>
          <input
            type="radio"
            value={opt}
            checked={selected === opt}
            onChange={() => setSelected(opt)}
          />
          {opt}
        </label>
      </li>
    ))}
  </ul>
  <button onClick={handleSubmit}>Next</button>
</div>
);
};

```

export default Quiz;

• SHOW RESULT AND RESTART (Result.js)

```

import React from "react";

const Result = ({ score, total, onRestart }) => {
  return (
    <div className="result">
      <h2>Quiz Completed!</h2>
      <p>Your Score: {score} / {total}</p>
      <button onClick={onRestart}>Restart Quiz</button>
    </div>
  );
};

```

export default Result;

• BASIC STYLING(Optional)

```

.app {
  text-align: center;

```

```
font-family: sans-serif;
padding: 2rem;
}
.quiz, .result {
margin-top: 1rem;
}
button {
margin-top: 1rem;
padding: 0.5rem 1rem;
}
```

- BASIC STYLING(optional)

```
.app {
text-align: center;
font-family: sans-serif;
padding: 2rem;
}
.quiz, .result {
margin-top: 1rem;
}
button {
margin-top: 1rem;
padding: 0.5rem 1rem;
}
```